Fruit Monitoring and Harvest Date Prediction Using On-Tree Automatic Image Tracking

Jaime Giménez-Gallego, Jesús Martínez-del-Rincon, Pedro J. Blaya-Ros, Honorio Navarro-Hellín, Pedro J. Navarro, and Roque Torres-Sánchez

***Abstract*—Fruit harvest date prediction is crucial to optimize resource management, maximize quality, and minimize waste of this food. For this purpose, it is necessary to monitor the fruit ripening stage. However, current measurement procedures pose drawbacks for widespread field deployment: laboratory trials are manual, destructive and expensive; measurements with hand-held portable equipment in the field are very time consuming; and the use of remote sensing mobile platforms has a high operating cost. In this article, a low-cost autonomous fixed sensor for continuous on-tree monitoring of pomegranates is proposed. It is based on a computer vision system able to extract reliable fruit color and size estimations automatically. In addition, an empirical quantitative and qualitative study on the effectiveness of using image-based monitoring in comparison with in situ manual and lab-based measurements for pomegranates is provided in this work. Another contribution of this article is a harvest date prediction model that employs the fruit information collected from the images. Furthermore, a thorough quantitative evaluation of the proposed prediction model for the fruit harvest date was performed, being the median error of the best model of 3.5 days.**

***Index Terms*—Artificial intelligence (AI), deep learning, fruit ripeness, fruit segmentation, internet of things, precision agriculture.**

I. Introduction

CLIMATE change poses a threat to the availability of water resources in arid and semi-arid areas. Concurrently, world population growth leads to an increase in agricultural commodity demand. This global scenario encourages research into the optimization of resources for agricultural production and minimization of food waste. Thus, one of the objectives in precision agriculture is the early prediction of harvest date, to facilitate planning for orderly harvest [1], so that production meets commercial quality specifications [2] and to reduce waste along the supply chain [2]. To this end, it is crucial to monitor crop status [3].

Numerous research works have addressed in the past the optimum scheduling of harvest by relying on the measurement of climatic parameters, especially ambient temperature; the date of full-bloom; and the use of machine learning based regression models over the recorded time series [1], [4], [5], [6]. Other articles have proposed the use of remote sensing through platforms such as drones [7], [8], [9] and satellites [10], [11], [12]. In the case of fruit trees, it is also very important to attend the fruit condition, as it provides the most direct information on the crop status. A relevant factor associated with fruit quality is its ripening stage. However, manual inspection by experts in field for grading ripeness is a time-consuming and error-prone method [13], [14]. Therefore, the development of automatic systems to determine fruit maturity on the tree represents an important advance in the applications of precision agriculture.

Multiple investigations have been conducted on the measurement, under laboratory conditions, of fruit ripeness and other quality parameters in various types of crops [2], [14], [15], [16], [17], [18]. Nevertheless, these measurements are nonautomatic, destructive and require the transportation of samples from the field to the measurement site, which is a logistical disadvantage. In paper [19], a system was designed to be mounted on a mobile platform that allows fruit pregrading during harvesting. This is an in-field measurement, but it is destructive and not intended to the periodic control of the crop status. The same problem is presented by the device proposed in [20], aimed at estimating the quality of fruits in bins at harvest. There are also portable equipment that allow direct non-destructive measurements to be taken in situ [14], [21], such as calipers, RGB cameras, colorimeters, DA-meters, spectrometers, spectral imaging devices and acoustic impulse resonance frequency sensors [2]. The drawbacks of these devices are that the measurement is still manual, on individual fruits and the information is captured from a small region of the fruit surface. Thus, it is necessary to perform several repetitions to ensure that measurements are representative in

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the case of fruits with nonhomogeneous appearance [14]. These reasons lead them to be very time-consuming.

Recent advances in computer vision techniques, supported by artificial intelligence (AI) tools, have made it possible to automatically detect and segment on-tree fruits [22], [23], [24]. Hence, despite the complicated field conditions, such as complex backgrounds, frequent fruit occlusions and natural light, several works have proposed to perform automatic nondestructive image-based in situ measurements of multiple fruits simultaneously [13], [22], [23], [25]. In some cases, the measurements were carried out from land [26], [27], [28] or aerial [25] mobile vehicles. Thus, a large area can be covered, which increases the volume of the fruit sample. Nonetheless, the cost of these platforms and of the specialized workers needed to operate them is still a constraint on the measurement process, limiting the amount of repetitions that could be performed during the ripening season.

Other studies, in contrast, used fixed measurement platforms [13], [23], [29], [30]. This strategy limits the number of monitored fruits but allows continuous and autonomous measurement. In addition, thanks to Internet of Things (IoT) sensors and protocols, measurements can be automatically communicated and remotely accessed, without the need to visit the field. These characteristics allow a complete analysis of crop development throughout the fruit ripening season. The image-based measurements performed in [13] and [23] were oriented to fruit size estimation. Fruit size is an interesting parameter, though by itself it does not provide complete information for harvesting decisions. For many fruits species, color can be considered the main visual indicator of ripeness. Furthermore, color is an important fruit quality factor that influences marketability and consumer preference [31]. Thus, gathering information on color is then essential for fruit monitoring. This could be achieved by means of RGB cameras [32], that have a very low cost. Hence, the installation of multiple devices would be feasible, mitigating the low spatial resolution problem of the fixed platform measurement. Nonetheless, the automatization of the process still remains challenging. For instance, fruit color development depends on direct sunlight, so less exposed fruits may achieve poorer coloration, causing over-maturity issues [33]. Moreover, in certain crops, color is not an adequate indicator of ripeness. For this reason, multispectral or hyperspectral cameras could also be employed [2]. However, the high cost of the equipment makes large-scale field deployment impossible, relegating it to the laboratory or mobile platforms. In [29], a mixed approach was conducted, tracking in image tomato flowers and fruits in the greenhouse to detect the flowering date, and predicting the harvest date using the daily average temperature. In [30], purple beans and tomatoes were monitored by imaging, soil moisture and temperature sensors, and environmental sensors; although the segmentation techniques utilized were not robust and did not allow individualized fruit analysis or counting.

In this article, we describe the development of a low-cost autonomous sensor to perform continuous on-tree monitoring of the pomegranate color and size evolution. The developed image-based sensors were installed in the field on a fixed measurement platform above the tree canopy with a zenithal perspective, as opposed to the lateral orientation adopted in [13] and [23]. In this way, field maintenance tasks were not hindered, and the continuity of the measurement time series was assured. The proposed cost-effective RGB sensor was evaluated by comparing over time its on-tree measurements with on-tree caliper and colorimeter measurements and laboratory colorimeter measurements. The aim of this study was also to obtain a model for the prediction of the pomegranates optimal harvest date using the time series based on the images captured by the sensors.

II. Materials and Methods

A. Field Measurement Infrastructure

1) Experimental Site: Trials were performed in a commercial orchard located in Mazarrón, Murcia, Spain (37°37'40.5"N 1°24'03.9"W). The measurement days took place along 2021 and 2022 during the fruit growth phase (from June to early September), the ripening phase (from early to late September) and the harvest (from late September to late October), which are crucial phenological stages for pomegranate trees (Punica granatum L. "Wonderful"). The trees were planted at a spacing of 5 m x 3.5 m (571 trees per hectare). A protective shade net was installed over the trees, which reduced the incident photosynthetically active radiation by 30% and evaporative demand. At the experimental site, nine sensors were deployed in order to acquire images of the fruits. In addition, approximately in the center of the experimental orchard, a Wi-Fi router over LTE was installed to enable remote communication of the camera measurements.

2) Sensor Description and Operation:

a) Sensor hardware and field set-up: A low-cost hardware platform was selected for field image capturing. The Raspberry Pi Zero W from Raspberry Pi Foundation (Station Road, Cambridge, U.K.), which has a Wi-Fi module, was used as the processing unit. For the RGB camera, the ZeroCam from Mann Enterprises LTD (Homefield Road, Haverhill, U.K.) with OV5647 sensor (5 MP, f2.9) was chosen. A 32 GB microSD card was used to store the files. The device was integrated by means of a 3-D printed bracket into a commercial IP67 ABS enclosure, where holes for the power cable and camera window were mechanized. As a protective measure against rainwater seepage, the power cable was connected through a gland cable, the hole for the camera was sealed using O-rings and a section of tube was glued as a barrier surrounding it.

The developed sensor was mounted pointing downward on a horizontal telescopic arm attached to one of the vertical metallic posts that support the crop protective net, as depicted in Fig. 1. The assembly allowed the length, the height above the ground, and the angle of the arm with respect to the vertical post to be adjusted. The sensor was located at approximately 1m from the crop canopy.

b) Sensor software, measurement protocol and communication: The operating system installed was Raspbian GNU/Linux 10 (buster) and the programming language used was Python 3.7.3. The picamera library [35] was used to take the images. The daily sensor procedure consisted of collecting

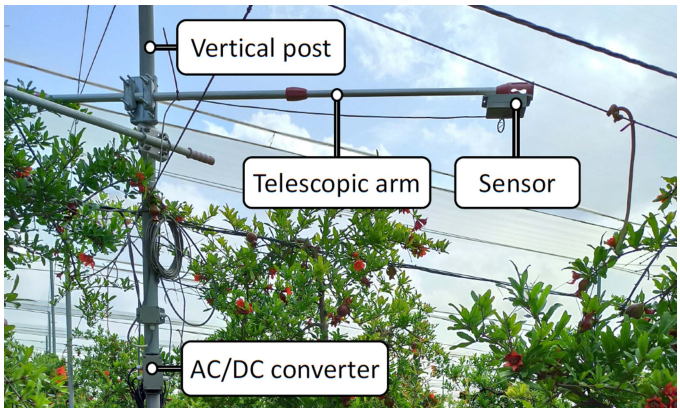


Fig. 1. Sensor set-up detail view [34].

five measurements, from 9 A.M. to 5 P.M. (GMT+2) every 2 h. The measurements were captured automatically using crontab, a Linux utility that allows to schedule tasks execution. Each measurement consisted of two repetitions, both spaced 1 min apart, to avoid the loss of information if the measurement coincided with an operator carrying out maintenance or any other eventuality that temporarily obstructed the camera's field of vision. For each repetition, an image was captured with the maximum resolution, i.e., 2592 × 1944 pixels.

Once the measurement was taken by the sensor, it was saved on the microSD card. In addition, to facilitate remote access to the data in real time and to avoid its loss in the case of the memory card breaking, the sensors were configured to automatically communicate the measurements via the internet through the router. For this purpose, a library was employed [36] which allows a folder on the Raspberry Pi to be synchronized with another folder hosted in the cloud, specifically the OneDrive platform was used. Thus, it was configured with crontab that, once the measure was finished, the synchronization of the local folder with the cloud folder was automatically executed, triggering the uploading of the new saved images.

c) Color correction: No fixed white balance settings were applied on the cameras used to record the images. Instead, the automatic mode was used, which oscillated between two white balance presets, one "cooler" and the other "warmer." The classification of the images into their corresponding group ("cool" or "warm") was easily automated based on the median value of the H channel of the HSV color space. Most of the pixel mass of the image was part of the tree canopy, therefore the median value of the H channel corresponded to a representative hue of green in the image. Thus, by a threshold for the median of the H channel (0.3), the image could be labelled as "cool" (above the threshold, bluer hue of green) or "warm" (below the threshold, yellower hue of green). To correct this difference and to be able to compare colorimetric measurements between images, a preprocessing was carried out using the editing software Adobe Lightroom. For each set of images, "cool" and "warm," based on an example of each group from only one of the cameras, we defined the values of the parameters that had to be adjusted to bring the images with different balances to a mid-point. For the

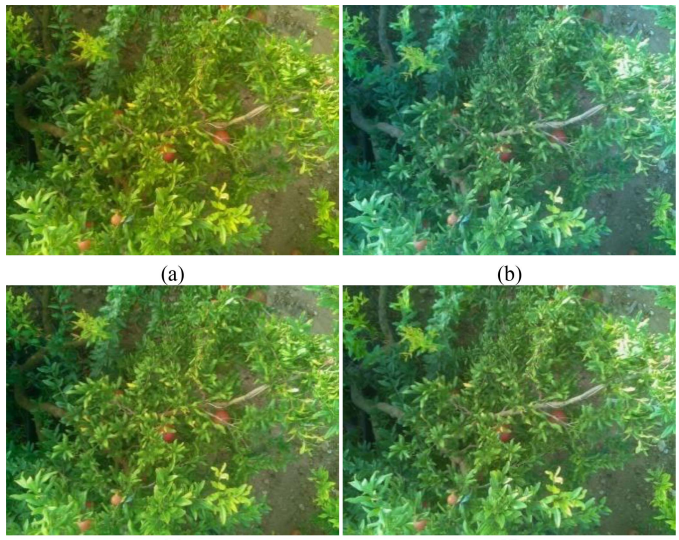


Fig. 2. Example images. (a) "Warm," 2022/09/26 9 A.M. (GMT+2). (b) "Cool," 2022/09/27 9 A.M. (GMT+2). (c) "Warm" corrected, 2022/09/26 9 A.M. (GMT+2). (d) "Cool" corrected, 2022/09/27 9 A.M. (GMT+2).

"warm" pictures: temperature –16K, hue +2 and exposure –0.1. For the "cold" pictures: temperature +21K, hue +5, exposure –0.5, green saturation +32 and green luminance +10. This correction was applied to all images from all cameras (see Fig. 2).

B. Comparison of Fruit Size and Color Measurements in-Field: Manual Versus Image-Based With Manual Segmentation

To justify the viability of image-based fruit monitoring, a comparison between direct manual fruit size and color measurements in the field versus those obtained from camera images with manual segmentation was first carried out. For both measurements to be comparable, they had to be carried out on the same fruits. Therefore, each year in July, after fruit thinning, the fruits of each monitored tree that were visible through the camera were marked with ribbons and an index was assigned to each of them. The fruits marked in 2021 and 2022 were 24 and 42, respectively.

Direct size measurements were taken between the months of August and October, in the slot time from 10 A.M. to 2 P.M., and were carried out on a total of five dates in 2021 and 7 in 2022. Fruit size was measured using a digital caliper (model 17–262, Acha, Eibar, Spain), with a resolution of 0.1 mm and an accuracy of ±0.2 mm. Specifically, the dimensions of the maximum and minimum diameters and the height of the pomegranate, defined from the base to the beginning of the crown, were measured and the mean was calculated. Direct color measurements were carried out in 2022, between September and October, in the slot time from 10 A.M. to 2 P.M., on a total of 4 dates. To measure fruit color, a PCE-CSM 2 colorimeter from PCE Instruments (Meschede, NRW, Germany) was used. Three repetitions were collected at different points in the region of the pomegranate visible by the camera. The mean was calculated for channels H

of the HSV color space (H_C) and a$*$ of the CIE L$*$a$*$b$*$ color space.

The image-based with manual segmentation measurements were carried out on the same dates as the corresponding direct measurements. In this case, for the fruit size, the bounding box (BB) of the fruit was manually annotated on the image and the average between the width and height was calculated. The image used to obtain the size on each date was the one shot at 9 A.M. (GMT+2), since the fruit was still present at that time on the harvest dates, allowing a greater number of data to be collected for comparison. Regarding the color measurement, it was conducted on all the images taken throughout the day: 9 A.M., 11 A.M., 1 P.M., 3 P.M. and 5 P.M. (GMT+2). Thus, five comparisons were obtained at different times with the direct manual measurements, in order to find out which case was the most interesting for a more accurate monitoring, given the influence of lighting in the camera color representation. To do this, first the fruit mask, i.e., a detailed binary mask containing all the pixel within the fruit silhouette and defined by the contour of that fruit, was manually annotated in each image. Then, in the region of interest (ROI) of the image, corresponding to the fruit mask, the median of the offset H channel (detailed below) of the HSV color space (H_m) and the a$*$ channel of the CIE L$*$a$*$b$*$ color space were calculated.

Regarding the H channel, it represents the color hue within the chromatic circle. However, when this circle is represented on a linear scale, an origin must be taken. As a consequence, two hues that are contiguous on the chromatic circle, but are separated by the origin, take on numerical values at opposite ends of the linear scale. This effect occurs in the red range by default, which is part of our colorimetric analysis range of interest, when the pomegranate fruit reaches an advanced stage of ripening. Therefore, so that proximity in the color hue would make mathematical sense throughout the entire range of the study, the origin of the numerical scale of the H channel was offset. Specifically, the origin was shifted to 0.5, the scale being from 0 to 1, now dividing the blue range, which does not affect the analysis of the color of the fruit. Thus, the range from green to red changed from 0.4–0 and 1–0.9, on the original H channel scale, to a range of 0.9–0.4, on the offset H channel scale. This way, the offset H channel is defined, which has been used from now on in the image processing of the cameras. This offset was not required for the a$*$ channel, since the a$*$ axis is relative to the green–magenta opponent colors, with negative values toward green and positive values toward magenta.

C. Time Series of Segmented Fruits in Images

To automatically monitor the different fruit characteristics from the image, a computer vision model was utilized to segment the individual fruit masks. For this purpose, a deep learning (DL) based AI model for instance segmentation called mask region-based convolutional neural network (MRCNN) [37] was used. This model was pretrained on the Microsoft common objects in context (COCO) dataset [38] and then finetuned with pomegranates examples to specialize in their segmentation. To this end, a set of 400 pomegranate tree images [39] was collected and manually annotated. The images had an original resolution of 4624 $\times$ 3472 pixels and were resized to reduce the computational cost of training and inference to a final resolution of 640x480 pixels. The images covered the full range of fruit ripening, being captured from June to October in 2021 and 2022. This dataset was divided into training (70%), validation (15%), and test set (15%). Data augmentation was used during training, including horizontal flip with probability 0.5 and Gaussian blur with probability 0.5 and random sigma between 0 and 0.5. Regarding the training parameters of the model, the stochastic gradient descent optimizer, a learning rate of 0.001, a learning momentum of 0.9, a regularization weight decay of 0.0001, and a batch size of 1 were selected. The training lasted for 80 epochs. Each epoch had 280 training iterations, one for each image in the training set. At the end of each epoch, the resulting model was evaluated in the validation set using the mean average precision (mAP) as monitoring metric. As training stop criteria, early stopping was defined to prevent overfitting, with a validation patience of 30 epochs.

The processing carried out in this article was performed on a computer with the following specifications: CPU Intel Core i5-8600K 6 cores @ 3.6 GHz, RAM 16 GB, GPU NVIDIA GeForce GTX 1070 Ti 8 GB GDDR5. The programming language used in this case was Python 3.6.13 and DL libraries were Keras 2.1.5, Tensorflow 1.15.0, and CUDA 10.0.

The trained instance segmentation model allowed to automatically obtain an independent mask for each fruit present in the image. However, please note that the assignment of the index to each mask does not follow a defined order. Moreover, fruits can vary their position, size and appearance across images and days, and some fruit may not be present in an image and then appear again. Therefore, over a sequence of images from the same camera, the mask corresponding to the same fruit may be assigned a different index in each image. To avoid this, we used a tracking algorithm [40] to assign and keep a unique identifier for each fruit across the days of year (DOY) and to determine, for each fruit, the time series of the masks generated by the MR-CNN model. The tracker configuration parameters applied were: "maxIters" 2, "linkEnergyThreshold" 100, "heightsPerSec" 1, "appearanceThreshold" 0.8, "nmsThreshold" 0.5, "maxFrameJump" 3, and "maxFrameJumpMultiplier" 7.

D. Automatic Image-Based Fruit Size and Color Measurements

To automatically monitor the evolution of the fruit size in pixels from the images, the masks of the fruit instances generated by the segmentation model were used. The pomegranate masks obtained had roughly a circular geometry without the crown. Hence, the radius of the circle fitted to the body of the pomegranate was defined as the characteristic parameter of the fruit size. Nevertheless, in most cases, the masks were incomplete due to occlusions caused by leaves, branches or other pomegranates ([see Fig. 5(g)–(i)]). This significantly increased the complexity of identifying the body of the pomegranate. There were protuberances on the masks that could be identified as either the crown or as part of the fruit body. Therefore, in order

to better estimate the real fruit size from the imperfect masks, the pomegranate mask was approximated to the circle that best fits the contour.

To automatically monitor the evolution of fruit color from the images, the masks of the instances generated by the segmentation model were also used. Thus, the ROI of each fruit was defined by applying the mask of the corresponding instance on the image. As characteristic parameters of the fruit color, H_m and the histogram of the offset H channel of the HSV color space were calculated in the ROI for each fruit. The histogram was calculated in the range from 0.4 to 0.8 with different numbers of bins (2, 4 and 8). Each bin count was normalized dividing by the fruit mask area, so the histogram was expressed as probability distribution. Hence, histograms of fruits whose masks did not have the same number of pixels would be comparable.

Once the methodology for automatically monitoring fruits from images was defined, its effectiveness was evaluated. To this end, a quantitative evaluation of fruit size and color measurements was carried out by comparing those estimated from images with automatic segmentation with those obtained from images with manual segmentation and with in-field colorimeter measurements (for color).

E. Comparison of Fruit Color Measurements: With Colorimeter in Laboratory Versus Automatic Image-Based in Field

For each camera mounted on a tree, the temporal sequence of the monitored fruits mean color (H_t) was calculated. It was obtained by averaging the color (H_m) of all the fruits present in the image on each date

$$H_t = 1 \Big/ N \sum_{i=1}^{N} H_m^i \tag{1}$$

where N is the total number of fruits tracked in a given image. Thus, a representative curve of the color evolution of the fruits of each tree was automatically obtained. At the same time, in 2021 and 2022 between mid-June and October, approximately every two weeks, fruit samples were collected. These samples were randomly gathered from the trees monitored by the cameras, although monitored fruits were never chosen, and the adjacent trees, exposed to the same irrigation strategy. On harvest dates, from the end of September to mid-October, a total of six fruits were picked as samples, while the rest of the dates, two fruits were collected as samples. The samples were taken to the laboratory, where color was measured with the colorimeter on the same day, taking three repetitions per fruit and calculating the mean (H_L). To demonstrate the usefulness of the developed system for monitoring the color evolution of a harvest sample, the automated camera measurements were compared with laboratory measurements.

F. Fruit Harvest Date Prediction

As an application of the developed tools for monitoring the size and color of pomegranate fruits, we propose to use this information to predict the harvest date on a per fruit basis. To this end, a machine learning (ML) based AI model for regression called XGBoost Regressor [41] was trained. The dataset used for training and testing was composed of the fruit time series obtained during 2 years with the nine cameras installed. Time series covering less than 40 days of fruit monitoring were filtered out. In total, the resulting dataset consisted of 68 fruit time series, which represented an average of approximately four fruits monitored per tree each year. Considering the daily measurements of each series, the dataset had 4850 data points within the time-series samples, which meant an average of 71 days per fruit. Each data point, which consists on daily fruit measurement, was divided into input and output. The input, as a feature vector (FV), contained information on fruit color, fruit size and date. The parameters of the full FV were: H_m, the normalized histograms of the offset H channel in the ROI with different number of bins (H_{hist2}, H_{hist4}, and H_{hist8}), the size in pixels and the number of elapsed days since an agronomic reference date (ED). The reference date used was the full-bloom date of pomegranate trees, which was obtained each season by counting a sample of trees on the experimental site: 19 May (2021) and 13 May (2022). Regarding the output of the regression model (y), the parameter used was the number of days remaining until the harvest date of the fruit. The harvest date of each fruit monitored by the cameras was manually annotated by checking the images and their timestamps.

The annotation of the fruit harvest date used for model training and for quantitative performance evaluation in testing was subject to some uncertainty, due to the harvest criteria used. Harvesting was carried out in a staggered manner, once a week from the end of September to mid-October. Each week, the fruits considered suitable according to the grower's quality standard, based on color, size, and shape, were harvested. This harvest criteria applied on the experimental site may not coincide in certain cases with the optimum criteria from an agronomic point of view. This was caused by logistical factors, since some fruits may have been gathered before the first harvest date, after the last harvest date, or on an intermediate date of those programmed for the plantation. There were also workers' subjectivity factors, since the same fruit, depending on the worker picking it, even if they followed common guidelines, could remain on the tree for one more or one less week. In addition, there were harvesting factors that were not visible in the image. Pomegranate fruits that were cracked, had not reached the minimum desired size, had not reached the desired color over the entire surface (including regions not visible on camera), or were deformed were left on the tree and removed on the last day. Therefore, on the last harvest day, fruits were collected in correct conditions, but also unmarketable according to the quality standard defined by the grower. Thus, for the latter, the annotated harvest date would be delayed with respect to what it should have been, judging by the information available from the model. Note that this can be considered as noise in the annotation process from the ML perspective that will affect both training and evaluation.

The fruit time series composing the dataset were randomly divided into training (70%), validation (15%), and test set (15%). For the prediction of fruit harvest date, the complete data time series was not used, since the objective of the regression model

TABLE I
FV CONFIGURATIONS

FV configuration	Features list	FV configuration	Features list
0	H_m	7	H_m, H_{hist4} and size
1	H_{hist2}	8	H_m, H_{hist4} and last ED
2	H_{hist4}	9	H_m, H_{hist4}, size and last ED
3	H_{hist8}	10	H_m, H_{hist8} and size
4	H_m and H_{hist2}	11	H_m, H_{hist8} and last ED
5	H_m and H_{hist4}	12	H_m, H_{hist8}, size and last ED
6	H_m and H_{hist8}		

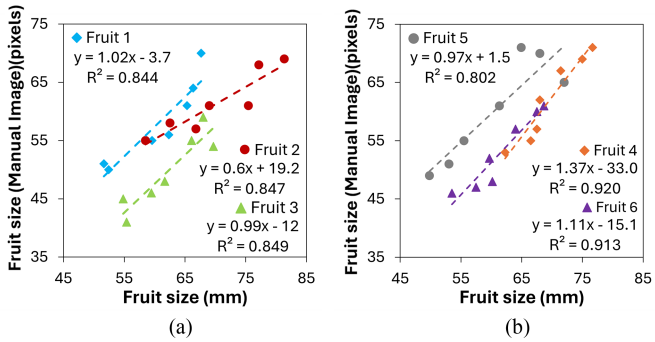


Fig. 3. Manual image-based versus caliper size measurements of field monitored fruits examples.

was to be able to predict in advance. Therefore, a time window was used that ran through the complete series generating subseries. Each data subseries was composed, as input, of the FV of each date included in the time window and, as output, of the remaining days until harvest with respect to the last date of the window. The time window size (TWS) parameter was defined as the number of days covered by the window, whose data were considered for the regression. Different sizes were experimented with: 1, 3, 5, 10, 20, 30, and 40 days. For each TWS, the data subseries were obtained and those with missing data (NaNs) were filtered out. This could happen punctually if the fruit was not detected in the image or was hidden by another element. For the remaining data subseries, different features were selected according to each of the defined FV configurations (see Table I). Thus, a final dataset was composed for each combination of TWS and FV configuration. By doing the training/validation/testing split before the subsampling, we ensure that the splits are not contaminated by having the same fruit in both training and testing, which will make results artificially high.

The XGBoost Regressor model was trained on each final dataset. Sets of values were defined for two of the main hyperparameters: the number of regression trees (25, 50, 100, 150, 150, 200, 250, and 300) and the size of those regression trees (from 2 to 14). For each combination of these parameters, the model was trained on the shuffled training set, while being evaluated on the validation set to monitor the training and perform early stopping. The metric chosen to measure regression performance was the root-mean-square error (RMSE). Training was stopped, to avoid overfitting, when ten epochs passed without a decrease in the RMSE of the model evaluated in the validation set. After completing the training, the optimal combination of number and size of regression trees was selected as the one in which the RMSE was minimized in the validation set. The model trained with this parameter configuration was evaluated in the test set. Thus, metrics were obtained that allowed comparing the accuracy in the prediction of fruit harvest date depending on the TWS and the FV configuration. The training process described above was repeated 150 iterations to avoid extracting biased conclusions due to a favorable distribution of the dataset within the training/validation/testing split.

III. RESULTS AND DISCUSSION

A. Comparison of Fruit Size and Color Measurements in-Field: Manual Versus Image-Based With Manual Segmentation

To evaluate the maximum accuracy achievable in monitoring fruit size using the cameras installed in the field, direct measurement with caliper (mm) was compared with manual image annotation (pixels), as described in the Materials and Methods Section II-B. This experiment made it possible to quantify the error derived from the evolution of the position of the fruit with respect to the camera over time. As ripening progresses, the fruit size increases and, consequently, its weight. Therefore, the hanging fruits could move away from the camera as the weeks pass by the effect of gravity. The greater the distance to the camera, the smaller the fruit size measurement in pixels. Thus, this effect would cause an increase in fruit size from the image to be unappreciated. Note that the image-based fruit size measurements were not comparable between different fruits, because they were located at different distances from the camera, which results on different scales or pixel sizes across fruits. Thus, only measurements of the same fruit over time could be correlated. For each of the marked fruits, where both measurements were taken on different dates in 2021 and 2022, the square of correlation coefficient (R^2) was calculated (see Fig. 3). The mean resulted in 0.8 and the median in 0.85. Therefore, based on the results, it could be considered that the evolution of fruit size was sufficiently represented in the image-based measurements.

Regarding fruit color monitoring by cameras, the direct measurement by colorimeter was compared with that obtained from the ROI of the manually annotated image, described in Materials and Methods. For each of the analyzed channels (channel a$*$ of the CIE L$*$a$*$b$*$ color space and channel H of the HSV color space) and for each time of day at which the image-based color measurement was obtained [9 A.M., 11 A.M., 1 P.M., 3 P.M., and 5 P.M. (GMT+2)], the R^2 between the measurements of all tagged fruits was calculated. The best results were found in channel H, with a mean R^2 for all hours of 0.7 and a maximum of 0.73. While in the a$*$ channel, a mean R^2 of 0.49 and a maximum of 0.54 were obtained. Furthermore, the measurements at all hours were analyzed jointly, as illustrated in Fig. 4, resulting R^2 of the H channel of 0.7 and that of the A channel of 0.51. Hence, for fruit color monitoring it was decided to use the H channel

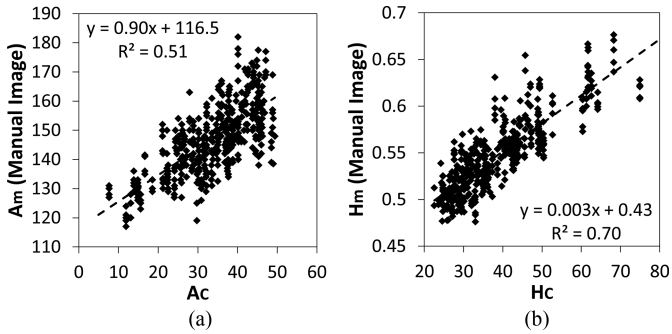


Fig. 4. Manual image-based vs. colorimeter color measurements of field monitored fruits at all hours. (a) A channel of the CIELAB color space. (b) H channel of the HSV color space [34].

TABLE II
FRUIT SEGMENTATION MODEL RESULTS

F1 score	Precision	Recall	mAP	IoU
0.817	0.799	0.836	0.671	0.784

of the HSV color space. Regarding the hours of measurement, the best results for the H channel were obtained for the images taken with the cameras at 9 A.M. (GMT+2), 0.72, and at 5 P.M. (GMT+2), 0.73. Although the difference with respect to the rest of the measurement hours is subtle, with the minimum R^2 being 0.67, this may be due to the lower presence of reflections on the fruits at early and late hours, as opposed to the central hours of the day. Although measurements at 5 P.M. (GMT+2) showed a slightly higher R^2, on the harvest dates the selected fruits had already been picked at that time. Therefore, in order to have more data for the fruit time series, 9 A.M. (GMT+2) was chosen as the measurement time. In view of the R^2 results, it could be justified that the image-based measurements were sensitive to the fruit color evolution in the ripening process. Moreover, it is expected that these results would improve with a wider range of fruit color measured. Field measurements with the colorimeter were taken between September and October, so the fruit was already at an advanced stage of ripening. Thus, the measurements were concentrated in a narrow color range. This impairs the correlation metric, as the ratio between the measurement range and the accuracy of the sensors is lower.

B. Fruit Image Segmentation

The results of the instance segmentation model evaluated in the test set, with a confidence threshold of 0.9, are presented in Table II. F1 score of 0.817 and intersection over union (IoU) of 0.784 were achieved for pomegranate. Other authors who had also used MRCNN for on-tree fruit segmentation obtained an F1 score of 0.868 and IoU of 0.878 [26], and F1 score of 0.86 [22], both for apple. The results obtained were considered sufficient for the intended application. A possible cause of error is the significant presence in the images of distant fruits, which would not be the monitoring target of the proposed sensor. These fruits have small pixel dimensions, so small absolute errors lead to substantial relative errors, affecting the mean of the metrics. Thus, it would be possible to improve fruit image segmentation results by filtering out fruits that were too far. To do so, it would be necessary to have depth data, captured by RGB-D cameras, light detection and ranging (LiDAR), or photogrammetry techniques.

C. Automatic Fruit Size and Color Time Series

As decided in Section III-A, daily images captured by the cameras at 9 A.M. (GMT+2) [42] were processed by the MR-CNN instance segmentation model to obtain the fruit masks. The tracker was then applied to these masks to obtain the mask index time series corresponding to each of the fruits. For each fruit, each day's mask was used to automatically collect size and color measurements, as described in Section II-D. Several examples of the results of the automatic image-based monitoring of the temporal evolution of fruit size and color are shown below (see Fig. 5, Videos 1–5 [43]).

Analyzing the graphs of the fruit size evolution time series, we can observe that the results are highly scattered in certain cases. This occurs throughout the whole time series in some fruits and in discrete stretches in other fruits. Size measurements that deviate from the general trend of the series are not reflected in a simultaneous analogous event in the fruit color time series. Occasionally, the segmentation model generates imperfect masks, which include regions from the background or miss regions from the fruit. These defects are transferred to the size estimation along with its own error. Nevertheless, color estimation is not as noticeably affected. Since it is not critical that the ROI is complete for the sample to be representative, nor that it includes undesired pixels since the median is a robust metric against outliers. On a qualitative level, it is appreciated that the fruit size automatically estimated from the image may be considered as a time series fruit monitoring parameter, if a high temporal resolution is used and outliers are properly filtered out. In this way, fruit growth trends can be reported, which is crucial to distinguish between crop phenological phases and, thus, to precisely adjust irrigation.

Regarding the graphs of the fruit color time series, it is observed that the curves follow the expected trends in a continuous way in most of the cases. It is possible to clearly distinguish the gradual process of color change on the fruit surface as the dates progress. This is essential to classify the crop ripening stage and, thus, to adjust irrigation and estimate harvest dates. Therefore, at a qualitative level, fruit color measured automatically from the image could be considered a useful parameter for time-series fruit monitoring. However, unexpected and momentary color changes are sometimes seen in parts of the time series of some fruits. These are mainly caused by inclusions of leaves and branches in the fruit masks, and to alterations in the orientation of the fruits by the operators. These cases illustrate punctual noise phenomena occurring during the automatic fruit color monitoring process. Furthermore, it is exposed that their origin does not exclusively lie in the optical sensors utilized for image capture or in their configuration, but also in the segmentation model and even in external interactions with the fruits.

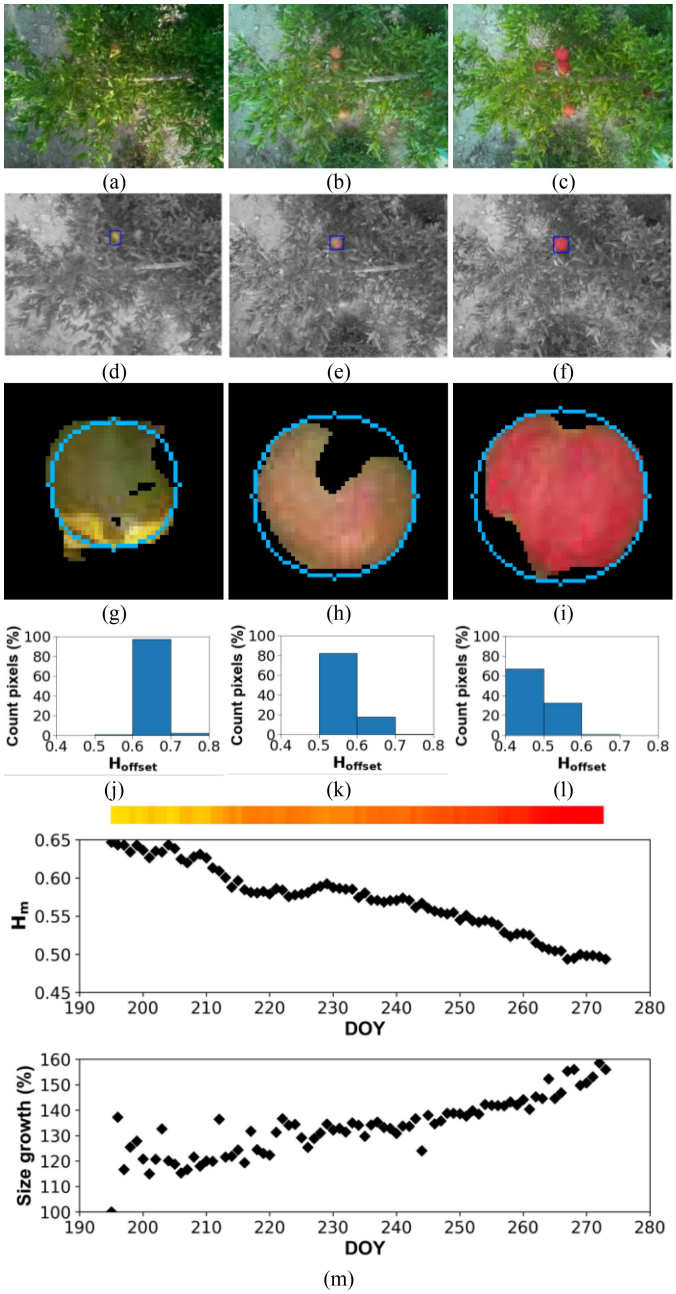


Fig. 5. Example of automatic image-based fruit monitoring (sensor 6, year 2021): (a)–(c) RGB images, from left to right, DOY 195, 234, and 273. (d)–(f) Color masks of the monitored fruit overlaid on grayscale image and BB in blue, from left to right, DOY 195, 234, and 273. (g)–(i) Crops of ROI with the circle fitted to estimate fruit size in blue, from left to right, DOY 195, 234, and 273. (j)–(l) H_{hist4}, from left to right, DOY 195, 234, and 273. (m) From top to bottom, fruit color legend, H_m versus DOY and size growth (size relative to that obtained on the first date of the time series) versus DOY. Modified from [34].

D. Evaluation of Automatic Image-Based Fruit Size and Color Measurements

The automatic image-based fruit size estimation accumulates the errors derived from the segmentation model when generating the mask and from the mask circle fitting. These errors are added to the base error that was obtained through manual annotation of the fruit in the image. To quantify the error of the automatic method, the fruit size measurements in the image with automatic and manual segmentation were compared. As a result, an R^2 of 0.78 was achieved. In addition, in Table III, the absolute and relative error metrics are presented. In both cases, the mean error is much higher than the median, over 1.8 and 1.6 times, respectively. This indicates that the error distribution is clearly right-skewed. Furthermore, the standard deviation is of the same order of magnitude as the mean, indicating a high error dispersion. Therefore, as the mean is a statistical metric sensitive to extreme values, the median is considered to provide a more robust and representative estimation. The median relative error was 6.76%, while the median absolute error was 4 pixels. This shows that the error margins in the prediction are very narrow, with notable errors arising for a matter of a few pixels due to the small size of the fruits in the image. Error magnitude obtained in the size estimation could be assumable in comparison to what a human operator could do during harvesting. Based on the numerical results, it follows that automatic fruit size estimation in the image offers a wide error variability. Even so, it could be taken into account as a fruit monitoring parameter in time series, where the magnitude of punctual errors is not as relevant as the trends of the data.

TABLE III
ERRORS OF IMAGE FRUIT SIZE MEASUREMENT WITH AUTOMATIC VS. MANUAL SEGMENTATION

	Absolute error (pixels)	Relative error (%)
Mean	7.31	10.92
Median	4.02	6.76
Standard deviation	8.33	11.34

TABLE IV
ERRORS OF IMAGE FRUIT COLOR MEASUREMENT, H_{m}, WITH AUTOMATIC VERSUS MANUAL SEGMENTATION

	Absolute error	Relative error (%)
Mean	0.01	1.85
Median	0.006	1.07
Standard deviation	0.012	2.2

Regarding the automatic image-based fruit color measurement, the only source of error with respect to the image-based measurement with manual segmentation was the segmentation model. To analyze the error, the fruit color measurements in the image with automatic and manual segmentation were compared. This resulted in an R^2 of 0.8. In Table IV, the absolute and relative error metrics are presented. Again, the median is used as the most representative statistic. The median absolute error was 0.006, the color measurement full range being approximately 0.4, and the median relative error was 1.07%. Therefore, the results can be considered to be satisfactory and the error caused by the segmentation model to be acceptable.

The automatic image-based measurements were also compared with the direct in-field measurements taken with the colorimeter. The R^2 achieved was 0.68, which is similar to that of the direct versus image-based with manual segmentation measurements, 0.72 (see Section III-A). This result confirms the reduced error magnitude introduced by the automatic process.

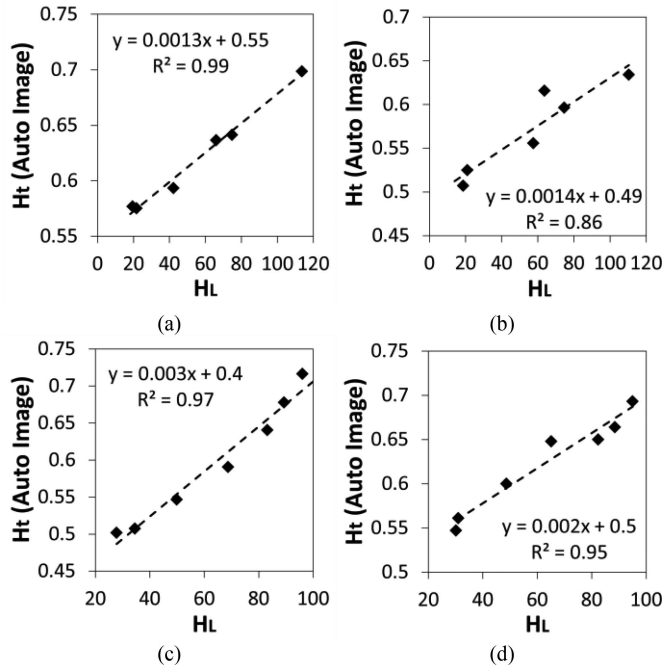


Fig. 6. Automatic image-based average color measurements of field monitored fruits, H_t, versus average color measurements with the colorimeter in the laboratory of harvested fruits, H_L, for four different trees (a)–(d) [34].

In addition, it demonstrates that the automatic color estimation from the image, proposed in this work, can be used for monitoring the fruit ripening status. As mentioned above, it is expected that these results would improve with a wider range of fruit color measured with the colorimeter.

E. Comparison of Fruit Color Measurements: With Colorimeter in Laboratory Versus Automatic Image-Based in Field

The automatic color measurements made on the fruits monitored by the cameras in the field, H_t, were compared with manual measurements with colorimeter in the laboratory on harvested fruit samples, H_L. This comparison can be only done at tree-level and not at fruit-by-fruit level, since the harvested samples during the ripening process were different fruits that the monitored ones, that for obvious reasons had to remain in the tree. For each season and sensor, for all dates when fruit samples were gathered and analyzed in the laboratory, comparative plots of the measurements were obtained (see Fig. 6). The calculated R^2 were very high, with a mean of 0.94, a minimum of 0.86, and a maximum of 0.99. Hence, a strong correlation was demonstrated between the two measurements for each monitored tree. The measures of all trees in the two years were also compared jointly. In this case, an R^2 of 0.79 was achieved. The color measurements compared, between the cameras in field and the colorimeter in laboratory, did not come from the same fruits and, in many cases, not even from the same tree. Therefore, these results demonstrate the capability of the developed sensor to monitor the color of a crop sample automatically and continuously, and to be able to filter small noise and errors at individual fruit level.

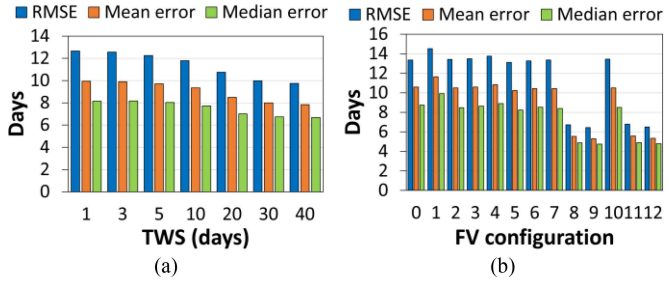


Fig. 7. (a) Average harvest date prediction results versus TWS. (b) Average harvest date prediction results versus FV configuration.

F. Fruit Harvest Date Prediction

To evaluate the harvest date prediction model, the following metrics were employed: RMSE, mean error, median error, standard deviation error, week accuracy (WA), fortnight accuracy (FA), and month accuracy (MA). The last three refer to the percentage of predictions whose error with respect to ground-truth (GT) remains within the correct week ($\pm$3.5 days), fortnight ($\pm$7.5), and month ($\pm$15 days), respectively. For example, in case the prediction was 25 days remaining to harvest and the GT was 20 days, it would not be considered within the correct week, but within the correct fortnight and month. These metrics were calculated in each of the 150 training iterations for each combination of TWS and FV configuration by evaluating the corresponding model in the test set. Subsequently, for each combination of TWS and FV configuration, the averages of the metrics for all iterations were calculated. In order to compare results between the different TWS, for each case the mean of all the FV configurations was calculated [see Fig. 7(a)]. Similarly, to analyze the performance of the model with the different features, for each FV configuration the mean for all TWS was determined [see Fig. 7(b)].

As depicted in Fig. 7(a), for all metrics, the larger the TWS, the better the average result. However, a smaller TWS reduces the days needed for prediction, enabling an earlier harvest date estimation. In addition, the smaller FV size leads to lower model weight and computational cost. Considering that the optimal parameter choice involves a trade-off between maximizing performance and minimizing TWS, the improvement from 20 to 40 days may not compensate. Regarding the comparison between FV configurations shown in Fig. 7(b), the best average results for all metrics were achieved with FV 9 and 12, which are configurations with all features included. Slightly worse results were given for FV 8 and 11, configurations in which fruit size is not taken into account. In a significantly worse performance range were the rest of the FV configurations, where ED is not involved. FV 7 and 10 were taken as representatives of this group of FV configurations as they presented similar results including a greater number of features. Therefore, from the results, it can be stated that the fruit size slightly improved the prediction result. Moreover, it is worth noting that including the ED feature improves the prediction of the model remarkably. FV 7, 8, and 9 obtained similar results to the corresponding FV 10, 11, and 12. The difference between both FV groups is that the color

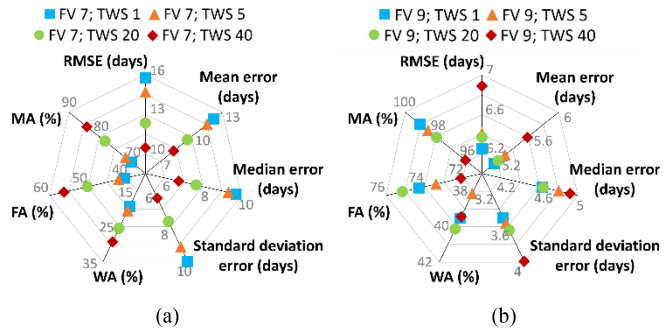


Fig. 8. Average harvest date prediction results for different TWS. (a) FV configuration 7. (b) FV configuration 9.

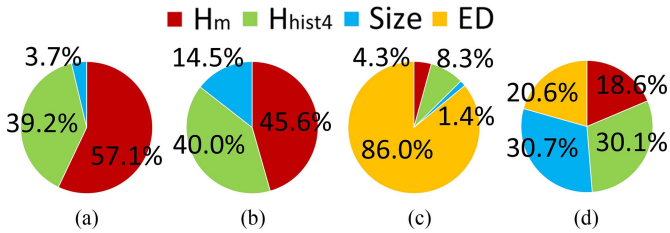


Fig. 9. Average feature importances. (a) FV 7 with TWS 1. (b) FV 7 with TWS 40. (c) FV 9 with TWS 1. (d) FV 9 with TWS 40.

histogram used in the first one counts with 4 bins and in the second one with 8 bins. Thus, FV 7, 8, and 9 were chosen as they have a smaller number of features.

Individual results of FV 7 and 9 for different TWS were compared. Fig. 8(a) shows the performance of the model with FV7. In light of the different metrics, it is clear that performance improves with larger TWS, as anticipated by the mean results discussed above. In contrast, as seen in Fig. 8(b), in the case of FV9 the results do not vary substantially with the TWS. A slight decline in the metrics can be discerned as the TWS increases.

Next, FV 7 and 9 were compared to each other with the best performing TWS in each case. FV 9 with a TWS of 1 day performed better on all metrics than FV 7 with a TWS of 40 days. Given only the evolution of fruit size and color with a time window of 40 days, on average, a median error of 6.74 days, and a MA of 84.03% were achieved. Adding temporal context to the regression model, a median error of 4.63 days and a MA of 99.04%. This demonstrates the importance for the regression model to have temporal context when predicting harvest date and how its inclusion reduces and/or eliminates the need of long time windows as input. Nevertheless, it is interesting that prediction results with FV 7, based exclusively on the fruit color and size evolution, do not differ greatly. Having a model that is not influenced by the number of days that have elapsed may have an application when forecasting the harvest in seasons with atypical climatology, crops subjected to irrigation treatments very different from the reference one, or species with different ripening pace.

In order to know the weight of the different features in the model decisions, the feature importances of FV 7 and 9 were calculated with various TWS, using "feature_importances_" model attribute of "Scikit-learn" library. Fig. 9 depicts that the importance of the feature "fruit size" increases the larger the

TABLE V
BEST HARVEST DATE PREDICTION RESULTS OF FV 7 WITH TWS 40 AND FV 9 WITH TWS 1

FV configuration	**9**	**7**
TWS	**1**	**40**
RMSE (days)	5.95	7.99
Mean error (days)	4.52	5.54
Median error (days)	3.46	3.76
Standard deviation error (days)	3.87	5.75
WA (%)	50.61	46.41
FA (%)	82.01	76.8
MA (%)	98.17	90.06

temporal window size, multiplying by 4 in the case of FV 7 and by 22 in the case of FV 9 when switching from a window size of 1 to 40 days. The importance of fruit color features is predominant for FV 7 in any case, and for FV 9 it is close to 50% with the largest TWS. Besides, in the case of FV 9, the importance of the ED feature decreases the larger the TWS, from 86% for a window size of 1 day to 20.6% for 40 days, which seems to indicate that the model is able to infer that information from longer time series.

To gather more information about the behavior of the regression model, importance distribution of the H_m parameter was also evaluated in function of the position within the time window. It was observed that the harvest prediction model focused more on fruit color in the later dates of the time window, regardless of the window size. This seems reasonable since this is the most up-to-date color information at the moment of prediction. It was also appreciated that in FV 9, where ED feature is considered, the relative importance of color in the dates before the end of the window was slightly higher.

The best training iteration was selected by taking the iteration in which average combination of TWS and FV was the best for the reference metric. FA was used as the metric of reference as a good compromise between functionality and accuracy performance. The results of the best training iteration, according to FA metric, for FV 7 with TWS 40 and FV 9 with TWS 1 are presented in Table V. As expected, the results obtained for the best iteration are better than those of the average of all iterations in practically all metrics. Furthermore, differences between the results for FV 7 and 9 are considerably lower than before. The median error with FV 7 is only 0.3 days larger than with FV 9, instead of the 2.11 days presented by the mean results. Another example is the FA, where the difference goes from 18.02% in the mean of the iterations to 5.21% in the best iteration. From the best results, we can see that our best model is able to effectively predict the harvest day with a mean error of only 4.5 days, a median error of 3.5 days, and within a fortnight ($\pm$7.5 days) in more than 80% of the cases. This seems therefore a promising tool for precision agriculture, especially if more data become available so that the model could be further improved.

As final analysis, the regression model predictions versus GT days to harvest of this training iteration were compared for FV 7 and 9 with TWS 1, 20, and 40. The R^2 was calculated in each case, and the mean value was 0.58 for FV 7 and 0.88 for FV 9. The correlation between errors and GT days to harvest was also computed for the same cases. It might seem logical that

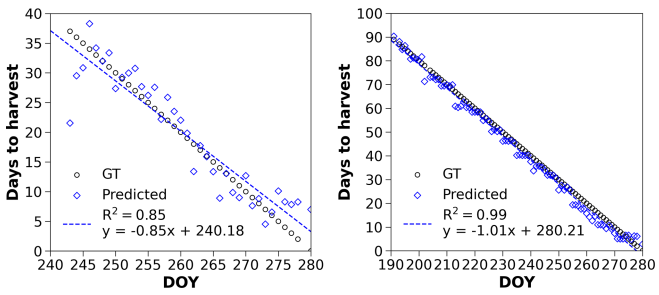


Fig. 10. Best training iteration predicted and GT days to harvest versus DOY. (a) FV 7 with TWS 40. (b) FV 9 with TWS 1.

the more days remaining to harvest, the greater the error in the model prediction would be. Nonetheless, the R^2 was very low in all cases, with a maximum of 0.26, indicating that there was no correlation. Time curves of predicted and GT days to harvest for a fruit of the test set are depicted in Fig. 10. From the graphs, it can be confirmed that the error is similar over the entire measured range. Consequently, it is concluded that the metrics presented above are representative independently of the days remaining to harvest, within the range studied in this article.

IV. Conclusion

In this article, a low-cost automatic image-based sensor for continuous monitoring of pomegranate fruit size and color was developed and field tested. Based on the results, it was shown that autonomous fruit size and color estimation from the image could be used to qualitatively know the evolution of fruit ripening. With this tool, fruit growth trends could be reported, and the crop ripening stage could be classified. This is crucial to accurately adjust irrigation, diagnose possible problems affecting the fruit ripening pace and estimate optimal harvest dates. In addition, the capacity of the presented sensor to automatically track the color of a crop sample in the field, in comparison with a commercial colorimeter in the laboratory, was demonstrated (R^2 of 0.79). Although visual indicators cannot replace the parameters obtained by destructive laboratory tests to measure the fruit quality, the proposed approach could be used as a complementary tool: a warning or decision support system to determine when the ripening stage is within a certain range of thresholds. Next, the main advantages of our solution compared to the use of a colorimeter, or any other device that requires in situ operation, are explained: First, the measurement time resolution of our sensor is unlimited. A higher measurement frequency does not increase the costs as it is an autonomous process, so it is not a constraint. Second, all measurements for multiple fruits can be performed simultaneously, regardless of the distance between measurement points, in contrast to manual measurement. Third, our system allows a scalable spatial resolution depending on the number of low-cost cameras installed. Instead, large-scale manned measurements would require an increase in the number of operators in addition to the measuring equipment. This would be accentuated in the case of multiple distant orchards. Finally, the measurement is performed remotely, nondestructively and is rapidly accessible.

As an additional contribution, from the automatic fruit measurements based on images captured by the sensors in the field, a harvest date prediction model for individual pomegranate fruits was designed and evaluated in this work. Our best model produces a median error of 3.5 days and a MA of 98% with only the need of a single day of automatic measurements. This also allows us to repeat the estimation every day for a continuous monitoring. The possibility of estimating in advance the approximate crop harvest date is very relevant for the logistic scheduling of the harvest, would allow an increase in fruit quality and improve post-harvest management.

The methodology proposed in this research would be applicable to other fruit and horticultural crops. Specially, those whose harvest criteria were closely related to the surface color evolution. Moreover, the image-based fruit monitoring strategy could be combined with classical methodology by adding climate information of interest such as ambient temperature, relative humidity, or solar radiation to the harvest date prediction model. It may also be possible to implement an AI model for flower detection to automatically obtain the full-bloom date. Although we have demonstrated in this article that the fruit monitoring can be performed with low-cost RGB sensors, the information available for the regression model could also be extended using multispectral cameras. Thus, fruit segmentation on the visible image could be utilized to accurately filter the layers of the different wavelengths of the multispectral image. In this regard, the features provided by the sensor developed in this article could be directly integrated into the thermal sensor proposed in [44]. This way, the ability to monitor fruit temperature would be incorporated.

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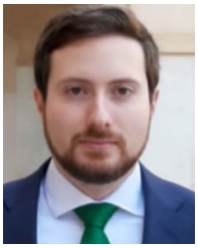
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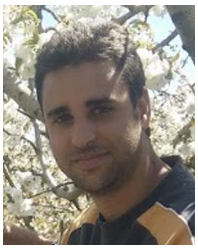
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